

Q1.1

(a) $L_{w,b}(x) = 0$

$$= (w_1, w_2) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + b = 0$$

$$w_1 x_1 + w_2 x_2 + b = 0$$

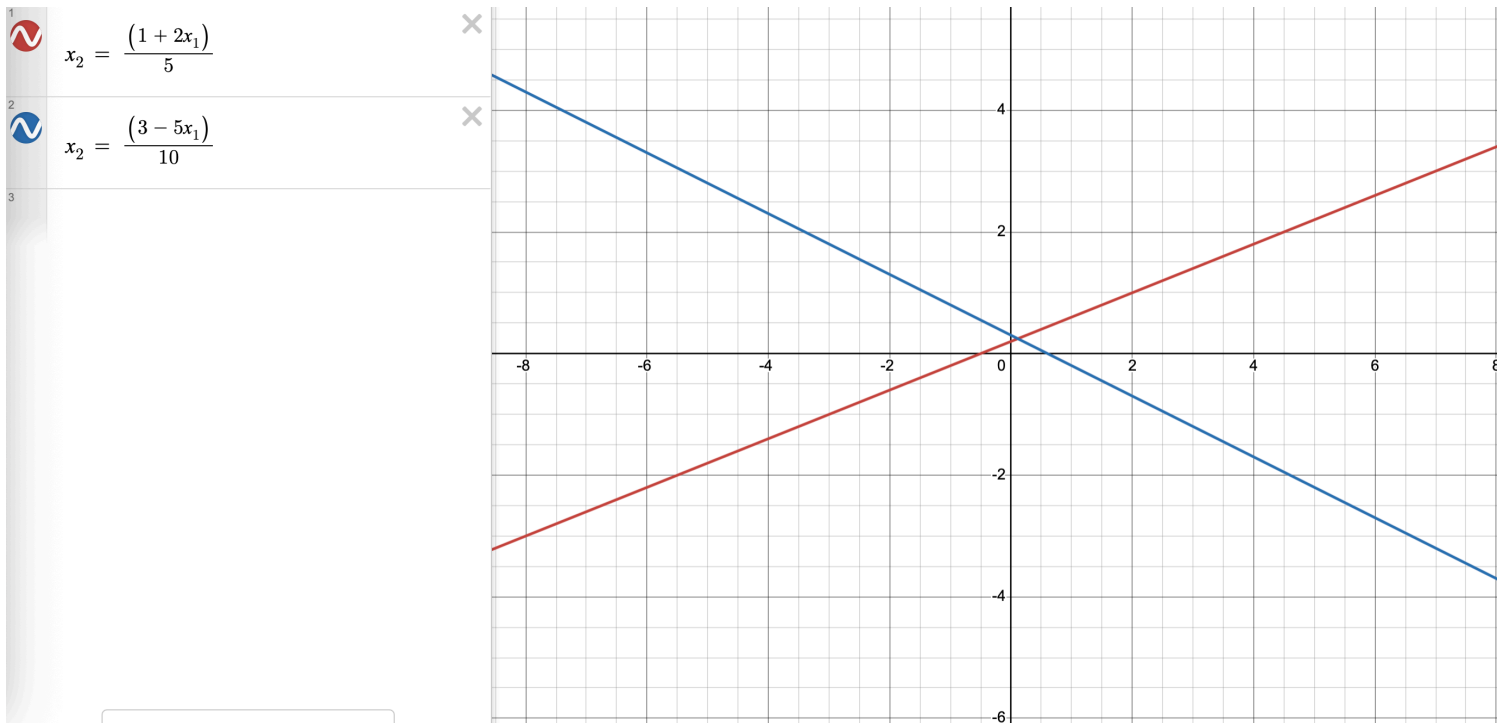
(b)

(i) $-2x_1 + 5x_2 - 1 = 0$

$$x_2 = \frac{1 + 2x_1}{5}$$

(ii) $5x_1 + 10x_2 - 3 = 0$

$$x_2 = \frac{3 - 5x_1}{10}$$



(c)

$$x = \text{ReLU}\{w[\text{ReLU}(wx+b)]+b\}$$

$$= \begin{pmatrix} 3 & -2 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

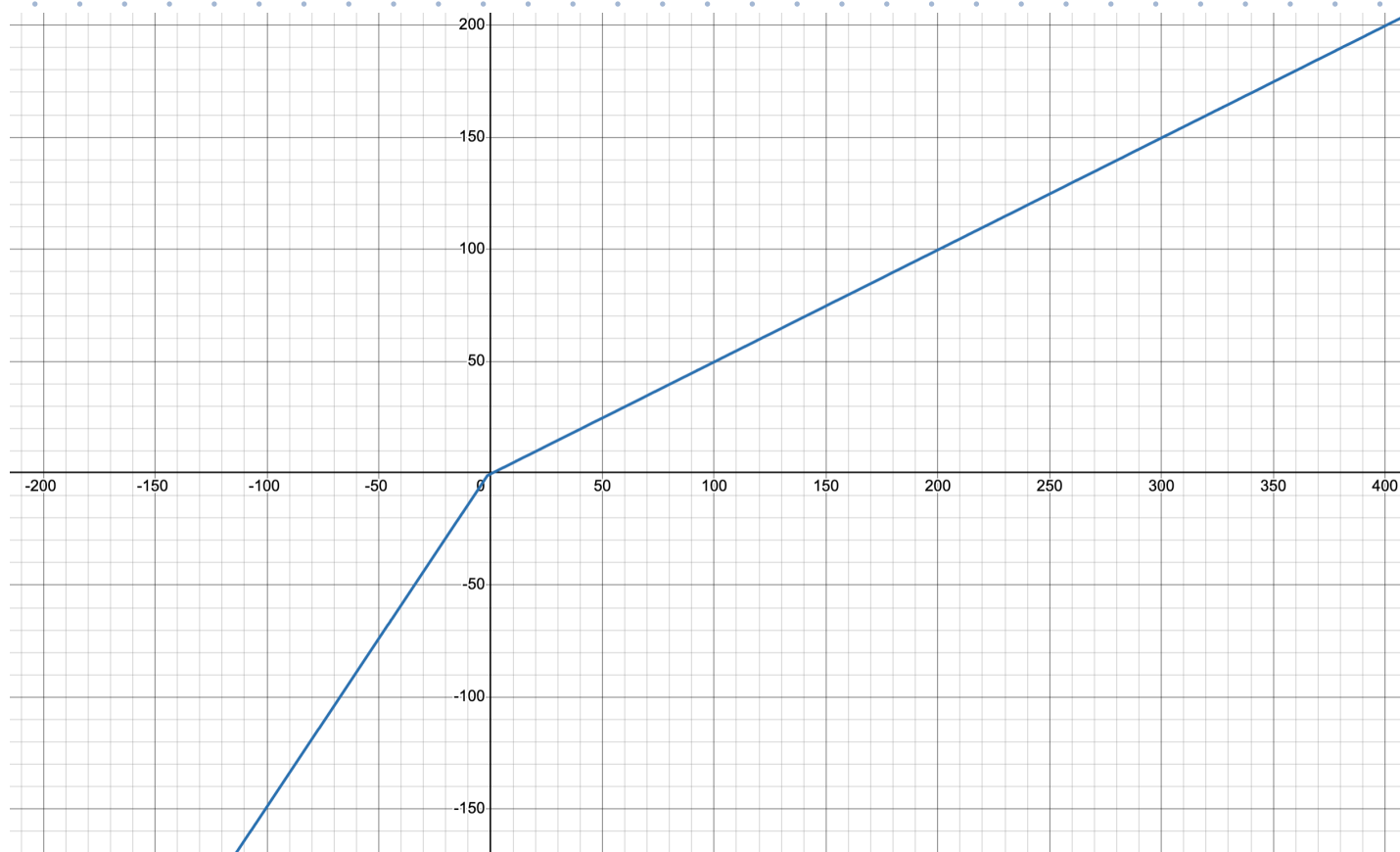
$$= \begin{pmatrix} 3x_1 - 2x_2 \\ -x_1 + 4x_2 \end{pmatrix} + \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

$$= \begin{pmatrix} 3x_1 - 2x_2 + 2 \\ -x_1 + 4x_2 - 3 \end{pmatrix}$$

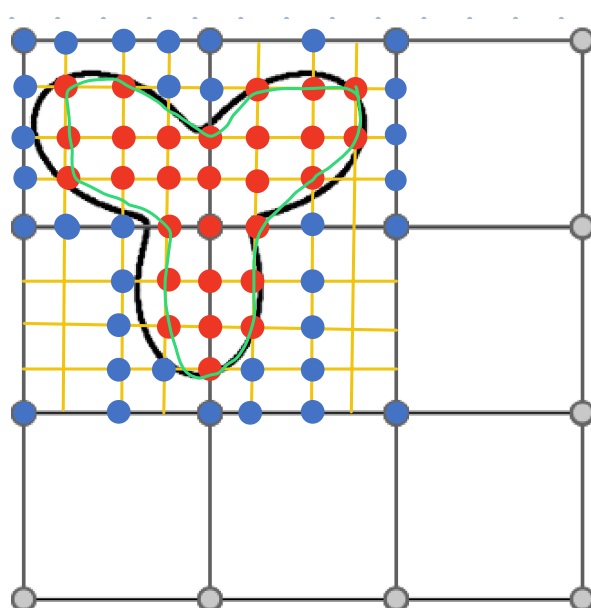
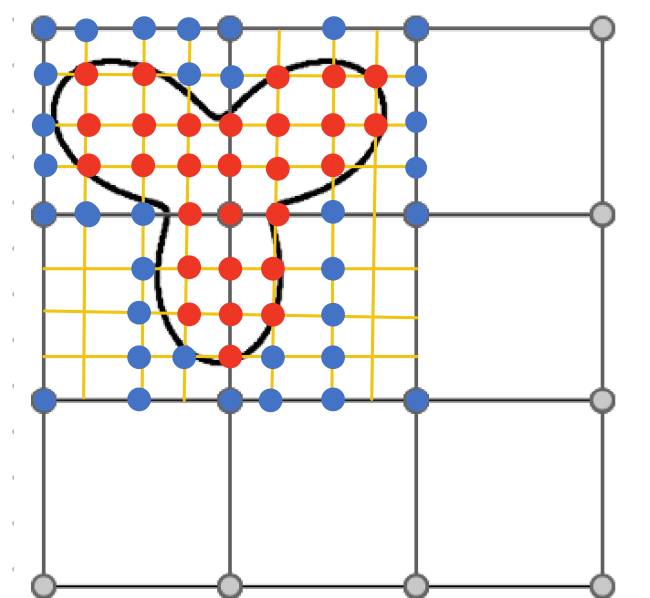
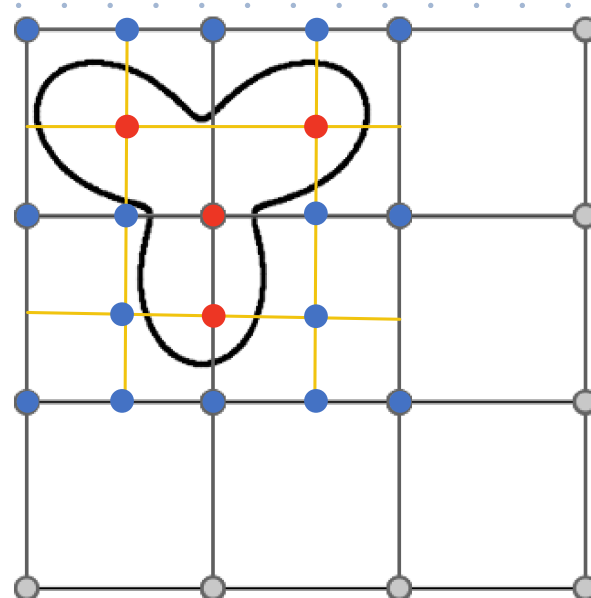
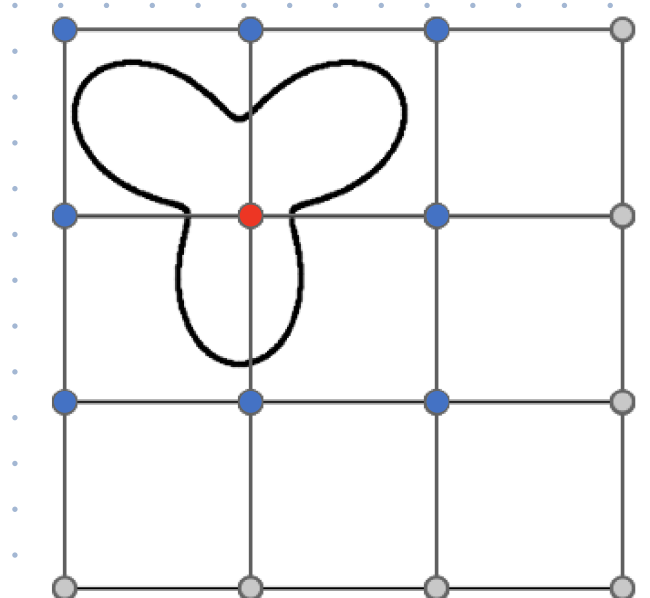
$$(1, -2) \begin{bmatrix} \max(3x_1 - 2x_2 + 2, 0) \\ \max(-x_1 + 4x_2 - 3, 0) \end{bmatrix} + (-1)$$

$$\max(3x_1 - 2x_2 + 2, 0) - 2 \max(-x_1 + 4x_2 - 3, 0) - 1$$

$$\max(\max(3x_1 - 2x_2 + 2, 0) - 2 \max(-x_1 + 4x_2 - 3, 0) - 1, 0)$$



(d)



Q1.3

(a)

(i) Linear (1024, 256)

input: [B, 1024] out: [B, 256]

$$W = [1024, 256], b = [256]$$

$$P\# = 1024 \times 256 + 256 \quad S_x = 1024 \\ = 262,400$$

(ii) Conv(16, 48, k=5, p=2, s=1)

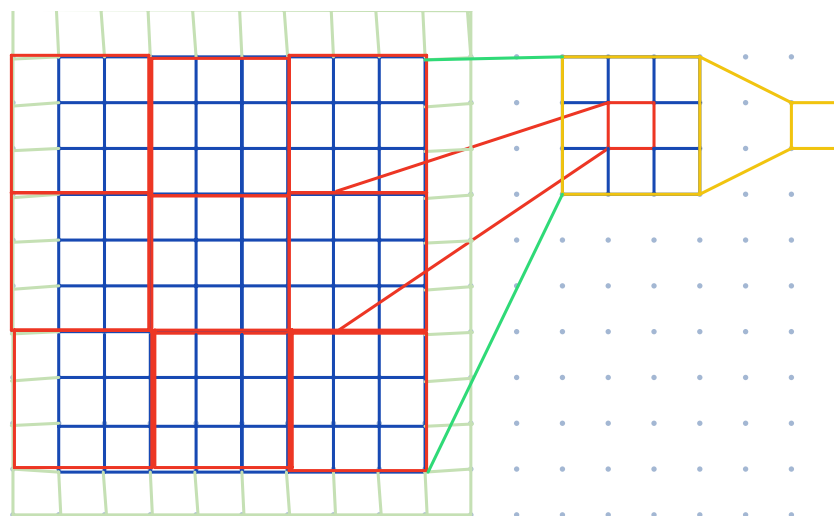
$$K = [48, 16, 5, 5] \quad \text{bias} = [48]$$

$$P\# = 48 \times (16 \times 5 \times 5 + 1) \quad S_1 = 5 \times 5 \\ = 19,248$$

(iii) Conv(64, 64, k=3, p=1, s=3, D=1)

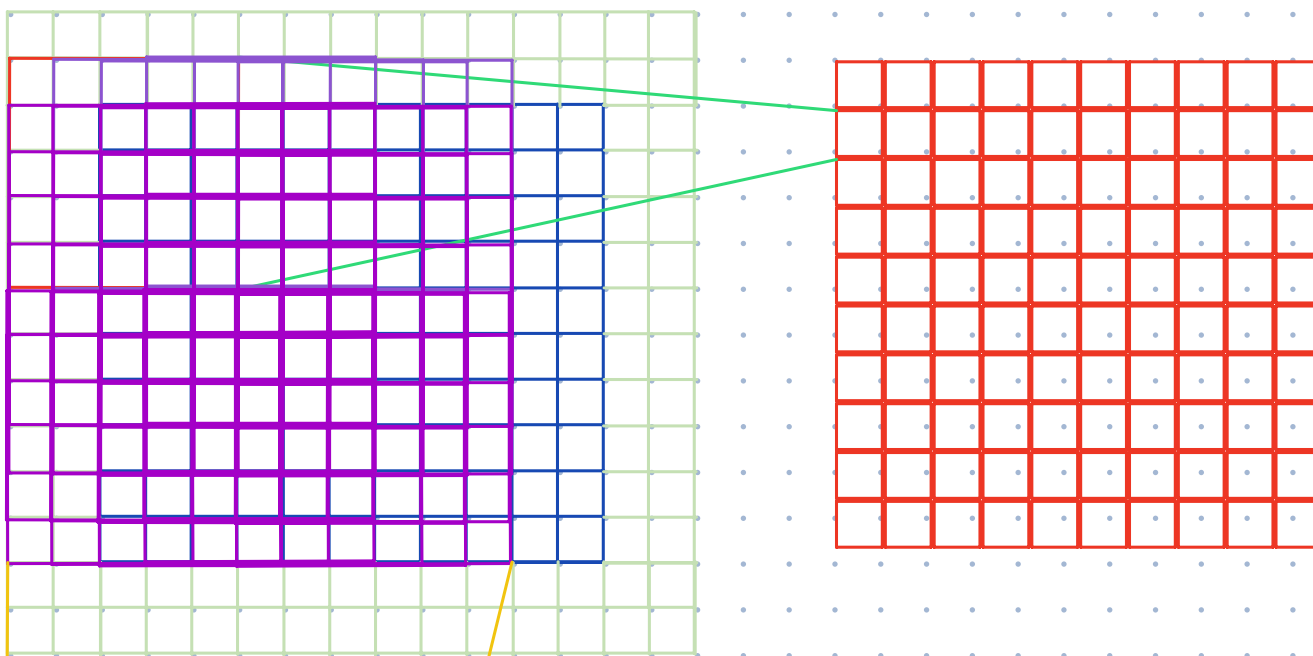
$$K = [64, 64, 3, 3] \quad \text{bias} = [64]$$

$$P\# = 64(64 \times 3 \times 3 + 1) \\ = 36,928 \times 2 \\ = 73,856$$

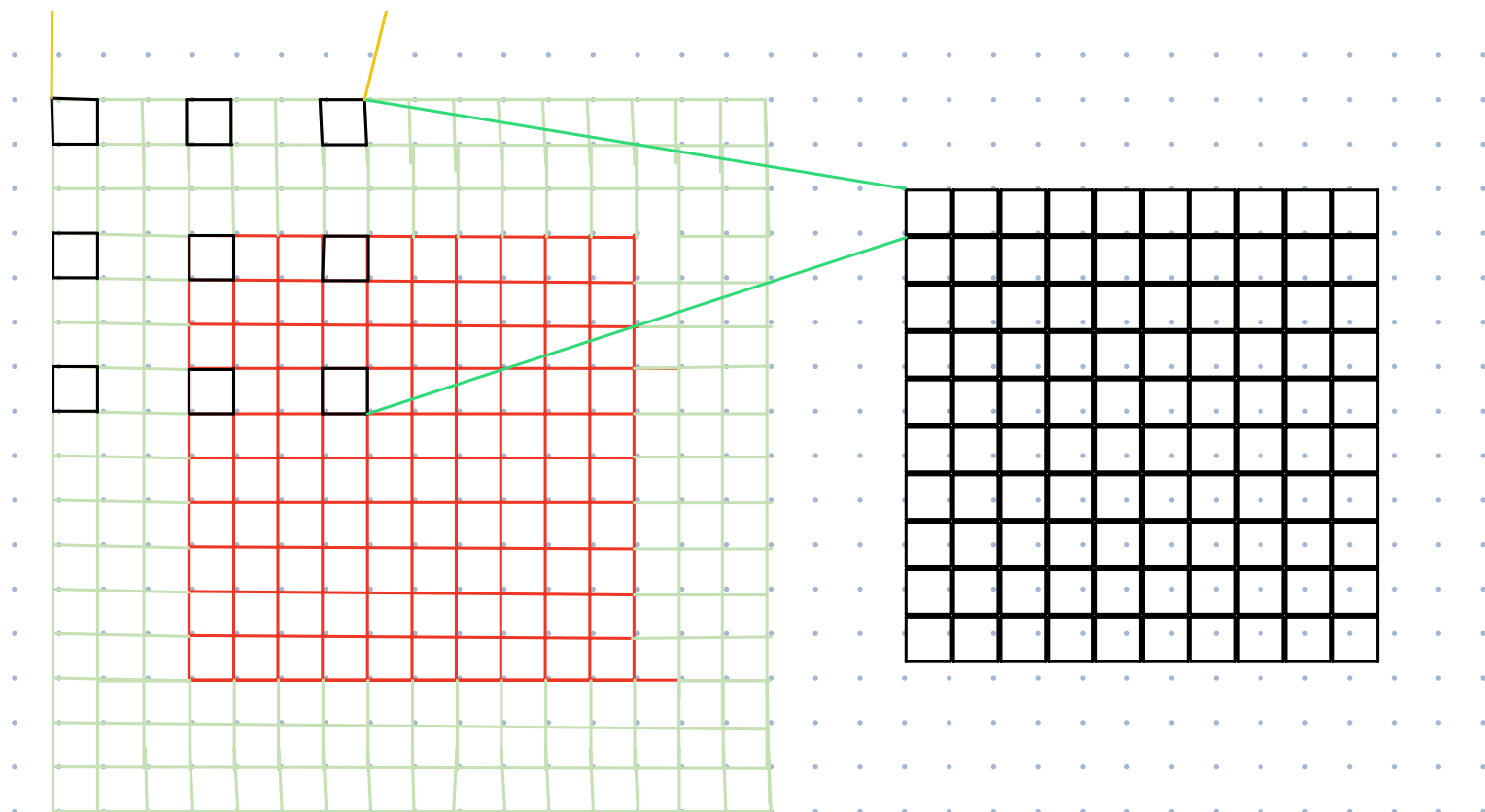


$$S_r = 9 \times 9$$

$$(\hat{N}) \text{ Conv}(24, 64, k=5, P=2, S=1, D=1)$$

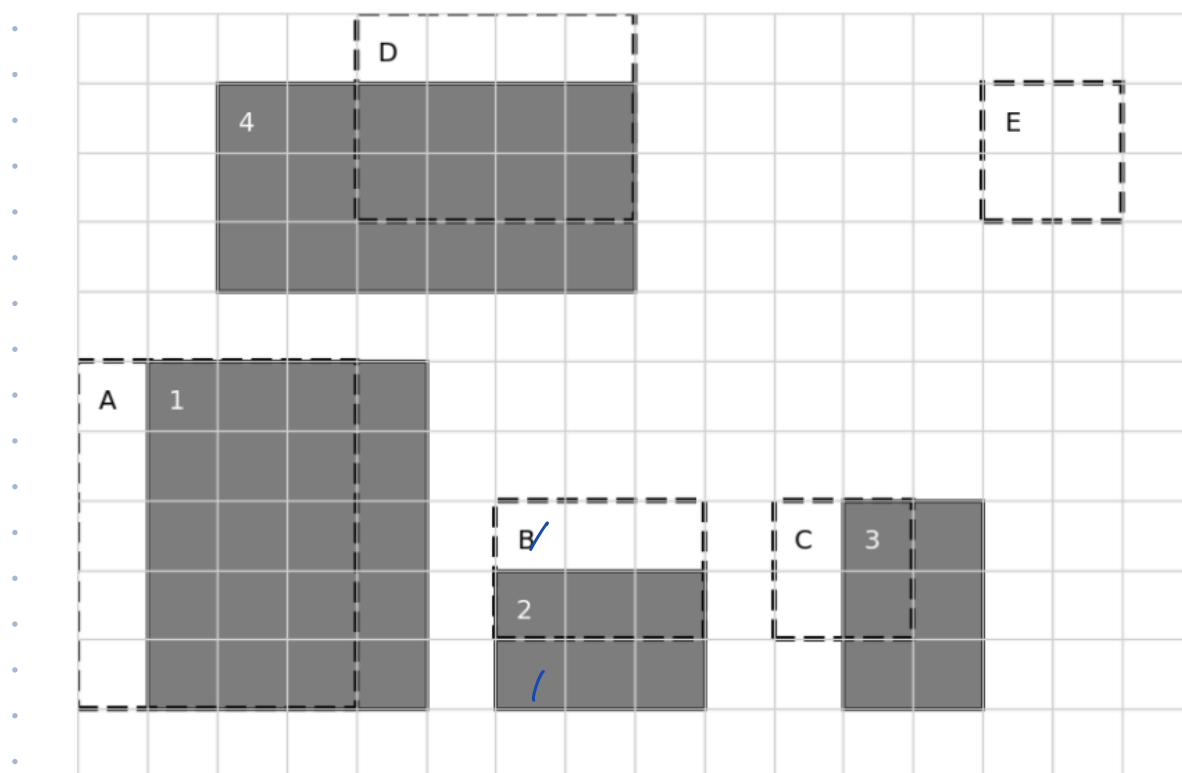


$$\text{Conv}(64, 128, k=3, P=3, S=1, D=3)$$



So, $S_r = 11 \times 11$

Q1.4



$$\text{IoU}(D, 4) = \frac{8}{22} = 0.36$$

$$\text{IoU}(A, 1) = \frac{15}{25} = 0.6$$

$$\text{IoU}(B, 2) = \frac{3}{9} = 0.333$$

$$\text{IoU}(C, 3) = \frac{2}{8} = 0.25$$

a) Correct detections @ 0.5 = 1 (c) TP = 3

(b) TP = 1

FP = 2

FN = 3

FN = 1

FP = 4

$$P = \frac{3}{3+2} = 0.6$$

$$R = \frac{3}{3+1} = 0.75$$

Q1.2

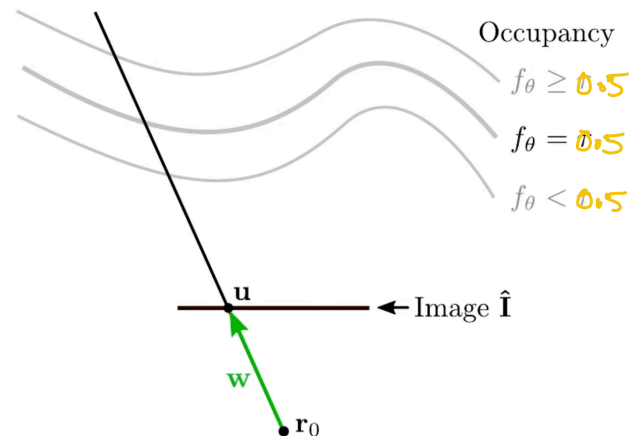
$$f_{\theta_f} = \sigma(Lw_f, b_f), \quad f_{\theta} = \tau, \quad \tau = 0.5$$

$$t_{\theta_t}(p) = \sigma(Lw_t, b_t)$$

Let,

$$W_f = (-3, 2), \quad b_f = 2$$

$$W_t = (1, 4), \quad b_t = -1$$



$$(a) \quad x = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + d \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}, \quad [d_0 = 1, d_1 = 2]$$

$$x_2 = x_1 - f(x_1) \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

$$\hat{p} = x_0 + \hat{d}w$$

$$P_0 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$P_1 = \begin{pmatrix} \sqrt{2} \\ \sqrt{2} \end{pmatrix}$$

$$f_{\theta_f}(P_0) = (-3, 2) \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} + 2$$

$$f_{\theta_f}(P_1) = (-3, 2) \begin{pmatrix} \sqrt{2} \\ \sqrt{2} \end{pmatrix} + 2$$

$$= 1.2928$$

$$= 0.58$$

$$= \frac{1}{1 + e^{-1.2928}}$$

$$= \frac{1}{1 + e^{-0.58}}$$

$$= 0.784 - 0.5$$

$$= 0.64 - 0.5$$

$$= 0.284$$

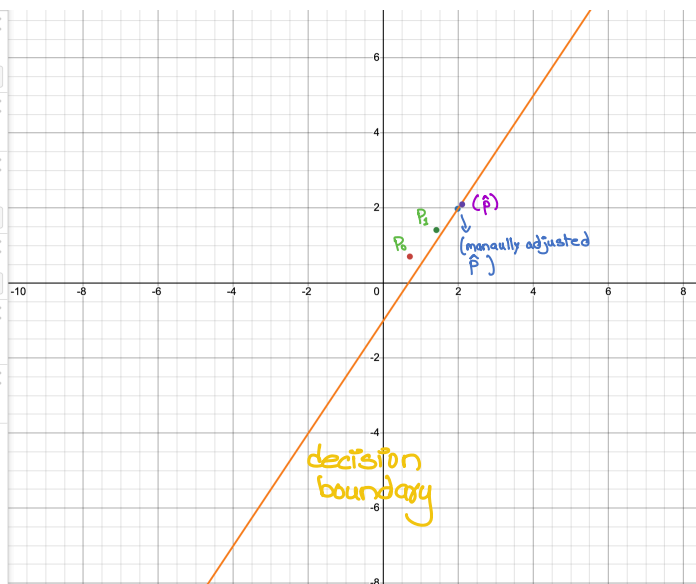
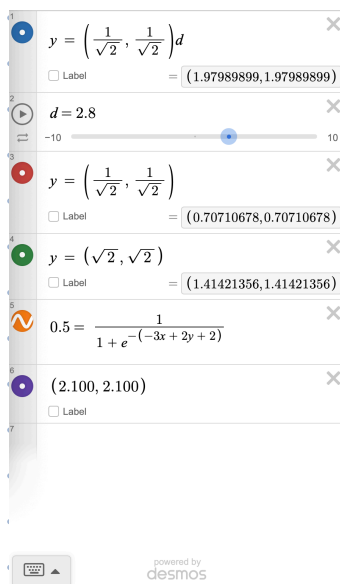
$$= 0.14$$

Now,

$$\hat{d} = \frac{2 - 0.14(2-1)}{0.14 - 0.284}$$

$$= 2.97$$

$$\text{So, } \hat{p} = \begin{pmatrix} 2.100 \\ 2.100 \end{pmatrix}$$



$$(b) f_{\theta_k}(\hat{p}) = (1, 4) \begin{pmatrix} 2.100 \\ 2.100 \end{pmatrix} - 1$$

$$= 9.5$$

$$= \frac{1}{1 + e^{-9.5}}$$

$$= 0.99$$

(c)

$$L = (\hat{I} - I)^2$$

$$\frac{\partial L}{\partial \hat{I}} = 2(\hat{I} - I)(1 - 0)$$

$$= 2(\hat{I} - I)$$

$$= 2(0.99 - 0.7)$$

$$= 0.58$$

$$\frac{\partial t_{\theta_t}(\hat{p})}{\partial w_f} = \left[\frac{\partial t_{\theta_t}(\hat{p})}{\partial \theta} + \frac{\partial t_{\theta_t}(\hat{p})}{\partial \hat{p}} \frac{\partial \hat{p}}{\partial \theta} \right]$$

$$\frac{\partial t_{\theta_t}(\hat{p})}{\partial w_f} = (0, 0)$$

$$t_{\theta_t}(\hat{p}) = \sigma(2 \cdot w_f \cdot b_t)$$

$$\frac{\partial t_{\theta_t}(\hat{p})}{\partial b_f} = 0$$

$$\frac{\partial t_{\theta_t}(\hat{p})}{\partial \hat{p}} = \sigma(w_t \hat{p} + b_t)$$

$$= \sigma(w_t \hat{p} + b_t) (1 - \sigma(w_t \hat{p} + b_t)) (w_f)$$

$$= 0.99 (1 - 0.99) (1.4)$$

$$= (0.0001, 0.0004)$$

$$(d) \quad \hat{p} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \hat{d} \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$